

# When a Wrist Frame Needs a New Observation

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

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## Abstract

Paper P studies a donor wrist as a handover frame when the held object is hidden, while paper Q surveys how predictions should be judged by available information, task error and cost. The thread asked whether the receiver's full visible history can still leave different object poses that call for incompatible grasps. The peers derived conditional decision and sensing bounds, but found no measured handover evidence for the premise. The verifier paused the thread because P already gives the basic obstruction and the added calculations do not yet support a distinct paper section. A matched-history experiment with receiving outcomes and sensing costs is needed to resume it.

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## 1 Introduction

Paper P is a local manuscript about handover when the object is occluded. It describes using the donor wrist pose as a substitute task frame. It also gives a conditional obstruction: if the receiver sees the same wrist, motor-encoder and action history in two trials, but the hidden object orientations need incompatible approaches, the receiver cannot choose correctly in both. P does not report the matched histories or receiving outcomes needed to show that this case occurs in practice [4].

Paper Q surveys learning-augmented algorithms, which use predictions while accounting for their errors. It says a deployed predictor can use only information available when it predicts. It also asks for an error measure tied to the task and for the cost of obtaining a prediction to be counted. Q is a survey, not a handover study [8].

The connection was a decision question. In-hand object rotation can change the object's pose relative to a fixed wrist. The peers asked whether the receiver's complete visible history then leaves a choice between object poses for which no single grasp works. They used Q's rules for observable predictions, task loss and query cost to make P's handover question more precise.

## 2 What was done

The peers first proposed treating the hidden object pose as a prediction target. They then moved from pose error to receiving success. A difference in pose matters only if it changes which actions can succeed. They wrote a common-grasp condition and a conditional failure

risk for a receiver that uses the full visible history. This followed P’s matched-observation argument but also covered cases in which a robust action succeeds across the possible poses.

They next checked a prediction-set certificate. If a covered set of poses admits one action that succeeds throughout the set, failure can occur only when the true pose falls outside it. One peer also considered a bound on grasp-point position error. The other pointed out that rotation about that point can leave its position unchanged while changing the required receiving orientation. They retained the full-pose, task-specific condition.

The peers then corrected two proposed transfers from Q. Q’s formal consistency claim still makes sense for either hidden state supplied with its own hypothetical exact prediction. The difficulty is deploying one predictor on identical observations in both states. They also found that a multiplicative cost ratio is ill suited to a one-shot handover with binary failure: a receiver told the true pose may have zero failure cost. They therefore used additive failure and sensing costs. Finally, they checked related work and sharpened the proposed physical test.

### 3 Results

Let  $Z = z$  be everything visible to the receiver at its decision, including wrist pose, encoder readings and action history. Let  $G$  be the hidden object pose relative to the wrist, and let  $S(G)$  be the set of receiving actions that succeed at that pose. For binary failure, the infimum risk over actions using only  $z$  is

$$R^*(z) = 1 - \sup_a \Pr\{a \in S(G) \mid Z = z\}.$$

This is the lowest failure chance that such actions can attain or approach. The peers derived this form in the ledger and checked its two-state case. It expresses the issue in terms of receiving decisions, rather than pose spread alone.

For a guarantee across all poses compatible with  $z$ , a successful history-only action exists exactly when their sets  $S(G)$  have a common member. P already states the obstruction for two matched states that require incompatible approaches [4]. The positive condition explains why pose variation alone need not cause failure: a common safe grasp would settle the choice.

The peers also checked a limited prediction-set result. Suppose a pose set computed from  $Z$  contains the true  $G$  with marginal probability at least  $1 - \alpha$ , where  $\alpha$  is the allowed miss probability. If the receiver chooses an action that succeeds for every pose in that set, and the object remains fixed until the action, its marginal failure probability is at most  $\alpha$ . This follows because failure implies that the true pose was outside the set. It says nothing about safety for each particular history.

For a simple sensing calculation, the peers assumed two equally likely, fixed hidden poses with nonempty but disjoint successful-action sets. They called the state-dependent sensor reading  $W$ , and its two conditional laws  $K_0$  and  $K_1$ . Without that reading, the minimum failure probability is  $1/2$ . With it, the minimum is  $(1 - \text{TV}(K_0, K_1))/2$ , where TV is the total variation distance between those laws. If failure costs  $M$  and sensing costs  $\kappa$ , with no other costs, sensing reduces expected additive cost exactly when  $\kappa < M \text{TV}(K_0, K_1)/2$ . The peers checked this binary decision calculation independently. Repeating a predictor on the same history with randomness independent of the hidden state supplies no new state information; the reading must depend on the hidden state. This is a conditional calculation, not a measured handover result or a direct use of Q’s predictor-call ratio [8].

## 4 What did not hold

An early suggestion said Q's consistency condition was vacuous on matched states. The peers withdrew that wording: Q tests consistency on an instance paired with an exact prediction, while the handover limit concerns whether one deployed predictor can produce distinct exact poses from the same input. They also rejected grasp-point position error alone as a certificate for an orientation-sensitive grasp. The one-shot failure model required additive accounting because its clairvoyant optimum can be zero.

The physical premise remained unsettled. P supplies no measured distribution of object pose given a complete wrist, encoder and action history. Approximate matches among continuous histories would need a justified tolerance and would not prove exact indistinguishability. A common safe grasp could remove the action conflict. A sensing action that moves the object would require a model of that change before the fixed-state calculation could be used.

The peers' prior-work checks also limited the claim. Work on uncertain-pose grasping with information gathering [2], conformal object-pose sets [6], and reported failures from object motion relative to a giver [5] already addresses nearby questions. They also noted robust grasps across uncertain shapes [3], moving-object handover [7], and haptic pose estimation under occlusion [1]. These checks do not establish the proposed matched-history effect for this pair, but they make the elementary certificate and sensing identity insufficient as stand-alone contributions.

## 5 Why the thread stopped

The verifier accepted the conditional-risk and sensing calculations under their stated assumptions. It paused the thread because P already states the basic matched-observation obstruction, while Q adds useful ways to account for information, error and cost without supplying a tested handover result. The smallest restart is to hold the donor wrist fixed, record the complete receiver-visible history, and externally measure the marked object's pose through controlled in-hand rotations or reseating. Candidate matched histories must then be checked for incompatible successful receiving actions. If they exist, held-out receiving trials can compare a history-only or common-grasp choice with one new observation, including its delay and cost. Until those measurements exist, the practical value of the connection is unknown.

## References

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